IN-CLASS QUIZ

Binary Heaps

Max-Heap Basics

4 multiple-choice questions · 1.5 minutes each

All heaps are max heaps, stored 1-based (A[1] is the root).

BEFORE WE START

Scan to answer

Open the form on your phone and keep it open for all four questions.

https://forms.gle/NsMt5CYiz8nY42F29

QUESTION 1

≈ 1.5 minutes

A max heap holds 100 distinct elements in A[1..100]. Taking a single node to have height 0, what is the heap's height, and at which indices could the minimum element be?

A Height 6; any index from 64 to 100

B Height 6; any index from 50 to 100

C Height 7; any index from 51 to 100

D Height 6; any index from 51 to 100

QUESTION 2

≈ 1.5 minutes

Insert 85 into the max heap A = [90, 70, 80, 40, 60, 50, 75, 20, 35, 55]. What is the resulting array?

A [90, 70, 80, 40, 85, 50, 75, 20, 35, 55, 60]

B [90, 85, 80, 40, 70, 50, 75, 20, 35, 55, 60]

C [90, 70, 85, 40, 60, 80, 75, 20, 35, 55, 50]

D [90, 85, 80, 40, 60, 50, 75, 20, 35, 55, 70]

QUESTION 3

≈ 1.5 minutes

Build a max heap from A = [10, 20, 30, 40, 50, 60, 70] using the bottom-up method (percolate down at each index from floor(n/2) down to 1). What is the resulting array?

A [70, 50, 60, 40, 20, 10, 30]

B [30, 50, 70, 40, 20, 60, 10]

C [70, 40, 60, 10, 30, 20, 50]

D [70, 50, 10, 40, 20, 60, 30]

QUESTION 4

≈ 1.5 minutes

What is remaining of the following max heap if you perform delete max twice: A = [90, 70, 80, 40, 60, 50, 75]?

A [70, 50, 80, 40, 60]

B [75, 70, 60, 50, 40]

C [75, 70, 50, 40, 60]

D [70, 40, 60, 50, 75]